

Event Detection Sensors: Research & Justification

All three sensors are read through a plain analog or digital pin. We deliberately avoid I2C/SPI parts so that wiring and debugging stay simple for a beginner team. Each is a low-cost breakout module with a large pool of Arduino/ESP32 tutorials.

1. Hazardous gas: MQ-2 gas sensor module

The MQ-2 is one of the most widely used and best documented gas sensors, has four pins (VCC, GND, AO, DO) and costs only a few dollars. It is a tin-dioxide (SnO_2) semiconductor whose conductivity rises as combustible gas or smoke increases. It responds to LPG, propane, methane, hydrogen, alcohol and smoke, over roughly 300-10,000 ppm (some vendors quote 200-10,000 ppm). The analog output (AO) swings roughly 0-5 V in proportion to concentration; the relationship is non-linear, so ppm values need the datasheet Rs/Ro curve and calibration. We read AO on an analog pin (the module's DO pin gives an optional hardware threshold via an onboard potentiometer). Practical notes: the heater needs 5 V (about 150 mA, under 900 mW), and the datasheet asks for a long first-use preheat (24-48 h), though modules settle enough for demos after a few minutes. On an ESP32, whose ADC accepts at most 3.3 V, AO should go through a two-resistor voltage divider, and an ADC1 pin should be used if Wi-Fi is on. Limitation to state in the report: the MQ-2 is not selective for toxic gases and is affected by temperature and humidity, so we use it as a threshold-above-baseline gas/smoke anomaly trigger, not as a calibrated toxic-gas measurement.

2. Structural collapse: SW-520D ball tilt/vibration switch module

We chose the rolling-ball SW-520D (not a mercury switch, which avoids a toxic part) on a small module with an LM393 comparator and a sensitivity potentiometer. It is the simplest possible collapse cue: no I2C, no library, no calibration. Inside the sealed switch a metal ball rolls between contacts, so it reports a binary state (tilted/not tilted, or a shock or knock) rather than a measurement. Listed specs: operates at 3.3-5 V, output current up to 15 mA, trips at roughly 14-45 degrees of tilt, open-circuit resistance above 10 MOhm, on-resistance under 5 Ohm, and a life of about 500,000 operations. Wiring: VCC, GND, and DO to any digital GPIO; the AO pin is unused. Vendors describe DO as high when the switch is open and low when closed, so the polarity should be checked on the actual module. Because the ball bounces, firmware should debounce and require a few triggers inside a short window. Limitation: it cannot tell how big an event was, and it also trips on the robot's own bumps.

3. Trapped worker: HC-SR501 PIR motion sensor

The HC-SR501 is the standard hobby PIR: three pins (VCC, OUT, GND), very cheap and extremely well documented. It senses changes in infrared radiation across its Fresnel lens, so it detects a warm body that moves, not one that is perfectly still. Detection range is adjustable by a potentiometer from about 3 m to a maximum of 7 m, over a cone of under about 120 degrees. A second potentiometer sets how long OUT stays high (about 0.3 s up to 5 min per the datasheet, other listings say 5-200 s), and there is a short re-trigger block time (0.2-2.5 s depending on the datasheet). It needs 5 V (accepts 4.5-20 V), draws up to about 65 mA, and needs 30-60 s of warm-up after power-on, during which triggers should be ignored. The output is a 3.3 V digital HIGH on motion, which is safe for ESP32 GPIO, so OUT goes to any digital pin. Limitation: the robot's own movement changes what the PIR sees, so readings are most reliable while the robot is stopped.

Sources: Winsen MQ-2 manual v1.4 (winsen-sensor.com); MQ-2 datasheet (Mouser); MQ-2 tutorials (PCBSync, Zbotic); uPesy ESP32 ADC guide (upesy.com); HC-SR501 datasheet (Rapid Electronics) and guides (ShillehTek, PIRHOME); SW-520D module listings (Eintronic, Temperosystems, Components101).